
DEPARTMENT OF MATHEMATICS

Mathematics Lab Assessment No. 3

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Question An automated PCB assembly line produces electronic control modules for industrial robots. The following data are observed:

Part A: Binomial Distribution

A batch contains 80 PCBs. Historical defect probability is 0.05.

Determine:

1. Expected number of defective boards.
2. Standard deviation.
3. Probability of exactly 4 defective boards.
4. Probability of more than 5 defective boards.

Part B: Poisson Distribution

A robotic welding cell experiences an average of 3.2 faults per shift.

Determine:

1. Probability of zero faults.
2. Probability of exactly 2 faults.
3. Probability of more than 5 faults.

Part C: Normal Distribution

Placement deviation follows:

$$X \sim N(0.50, 0.07^2)$$

Find:

1. Probability that deviation lies between 0.40 mm and 0.60 mm.
2. Probability deviation exceeds 0.65 mm.
3. 95th percentile deviation

Part D: Central Limit Theorem

Cycle times have:

$$\mu=15s, \sigma=2.4s \quad \mu=15s, \quad \sigma=2.4s$$

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	A sample of 36 cycles is selected. Determine: - Standard error. - Probability that sample mean exceeds 15.5 s. Part E: Correlation Analysis <table><thead><tr><th>Observation</th><th>Assembly Torque X (Nm)</th><th>Vibration Y (mm/s)</th></tr></thead><tbody><tr><td>1</td><td>11</td><td>2.0</td></tr><tr><td>2</td><td>14</td><td>2.6</td></tr><tr><td>3</td><td>13</td><td>2.4</td></tr><tr><td>4</td><td>17</td><td>3.3</td></tr><tr><td>5</td><td>15</td><td>2.9</td></tr><tr><td>6</td><td>18</td><td>3.6</td></tr></tbody></table> Determine: - Sample covariance. - Correlation coefficient. - Engineering interpretation.	Observation	Assembly Torque X (Nm)	Vibration Y (mm/s)	1	11	2.0	2	14	2.6	3	13	2.4	4	17	3.3	5	15	2.9	6	18	3.6
Observation	Assembly Torque X (Nm)	Vibration Y (mm/s)																				
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Solution	i) Identify the parameters and mathematical concept Parameters: - Binomial: $n = 80, p = 0.05$ - Poisson: $\lambda = 3.2$ - Normal: $\mu = 0.50, \sigma = 0.07$ - CLT: $\mu = 15, \sigma = 2.4, n = 36$ - Correlation: Torque X, Vibration Y Mathematical Concept: - Binomial distribution - Poisson distribution - Normal distribution - Central Limit Theorem (sampling distribution of mean) - Covariance and Pearson correlation coefficient																					

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	ii) Solve analytically
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Part A: Binomial distribution

Given $n=80$, $p=0.05$

$$\mu = np = 80(0.05) = 4$$

$$\sigma^2 = np(1-p)$$

$$= 80(0.05)(0.95) = 3.8$$

$$\sigma = 1.949$$

$$P(X=4) = {}^{80}C_4 (0.05)^4 (0.95)^{76}$$

$$P(X=4) = 0.2004$$

$$P(X>5) = 1 - P(X \leq 5)$$

$$P(X>5) = 0.2108$$

Part B: Poisson distribution

Given $\lambda = 3.2$

$$P(X=0) = e^{-3.2}$$

$$P(X=0) = 0.0408$$

$$P(X=2) = \frac{e^{-3.2} (3.2)^2}{2!}$$

$$= 0.2087$$

$$P(X>5) = 1 - P(X \leq 5)$$

$$P(X>5) = 0.1050$$

Part C: Normal distribution

Given $\mu = 0.50$, $\sigma = 0.07$

$$Z_1 = \frac{0.40 - 0.50}{0.07} = -1.43$$

$$Z_2 = \frac{0.60 - 0.50}{0.07} = 1.43$$

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$$P(-1.43 < Z < 1.43) = 0.847$$

$$= 84.7\%$$

above 0.65

$$Z = \frac{0.65 - 0.50}{0.07} = 2.14$$

$$P(Z > 2.14) = 0.0162$$

$$= 1.62\%$$

95th percentile

$$X = \mu + 1.645\sigma$$

$$X = 0.5 + 1.645(0.07)$$

$$X = 0.6152 \text{ mm}$$

Part D: Central Limit Theorem

given, $\mu = 15$, $\sigma = 2.4$, $n = 36$

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{2.4}{6} = 0.4$$

Mean exceeds 15.5

$$Z = \frac{15.5 - 15}{0.4} = 1.25$$

$$P(Z > 1.25) = 1 - 0.8944$$

$$= 0.1056$$

$$= 10.56\%$$

Part E: Covariance and correlation

$$\bar{X} = 14.67, \bar{Y} = 2.80$$

$$S_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1} = 1.852$$

$$S_x = 2.58, S_y = 0.59$$

$$r = \frac{S_{xy}}{S_x S_y} = 0.997$$

$$= 0.997$$

iii) GeoGebra Screenshot / Program Execution

```
import numpy as np
from scipy.stats import binom, poisson, norm

# Part A : Binomial Distribution
n = 80
p = 0.05

mean_bin = n*p
sd_bin = np.sqrt(n*p*(1-p))
prob_4 = binom.pmf(4,n,p)
prob_gt5 = 1-binom.cdf(5,n,p)

print("BINOMIAL")
print("Mean =",mean_bin)
print("SD =",sd_bin)
print("P(X=4) =",prob_4)
print("P(X>5) =",prob_gt5)

# Part B : Poisson Distribution
lam = 3.2

p0 = poisson.pmf(0,lam)
p2 = poisson.pmf(2,lam)
pgt5 = 1-poisson.cdf(5,lam)
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print("\nPOISSON")
print("P(X=0) =",p0)
print("P(X=2) =",p2)
print("P(X>5) =",pgt5)

# Part C : Normal Distribution
mu = 0.50
sigma = 0.07

prob_between = norm.cdf(0.60,mu,sigma)-norm.cdf(0.40,mu,sigma)
prob_above = 1-norm.cdf(0.65,mu,sigma)
percentile95 = norm.ppf(0.95,mu,sigma)

print("\nNORMAL")
print("P(0.40<X<0.60) =",prob_between)
print("P(X>0.65) =",prob_above)
print("95th Percentile =",percentile95)

# Part D : CLT
mu = 15
sigma = 2.4
n = 36
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SE = sigma/np.sqrt(n)
z = (15.5-mu)/SE
prob = 1-norm.cdf(z)

print("\nCLT")
print("SE =",SE)
print("P(Xbar>15.5) =",prob)

# Part E : Correlation
x = np.array([11,14,13,17,15,18])
y = np.array([2.0,2.6,2.4,3.3,2.9,3.6])

cov = np.cov(x,y)[0,1]
corr = np.corrcoef(x,y)[0,1]

print("\nCORRELATION")
print("Covariance =",cov)
print("Correlation =",corr)
```

iv) Results and analysis from the graph

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	BINOMIAL Mean = 4.0 SD = 1.9493588689617927 $P(X=4) = 0.2004311340447808$ $P(X>5) = 0.21077532149896283$ POISSON $P(X=0) = 0.04076220397836621$ $P(X=2) = 0.20870248436923497$ $P(X>5) = 0.10540810546917734$ NORMAL $P(0.40 < X < 0.60) = 0.8468725489803304$ $P(X > 0.65) = 0.016062285603828275$ 95th Percentile = 0.615139753886603 CLT SE = 0.39999999999999997 $P(\bar{X} > 15.5) = 0.10564977366685535$ CORRELATION Covariance = 1.5200000000000002 Correlation = 0.9979288897338915 • Binomial Distribution: Expected defects are low (4 out of 80 PCBs), indicating a stable manufacturing process. • Poisson Distribution: Fault occurrences are infrequent, showing good reliability of the robotic welding cell. • Normal Distribution: Most placement deviations fall within acceptable limits, indicating good quality control. • Central Limit Theorem: The probability of the average cycle time exceeding 15.5 s is low, showing consistent production performance. • Correlation Analysis: A strong positive correlation ($r = 0.85$) exists between torque and vibration; higher torque is associated with higher vibration, which can be used for predictive maintenance.